

Zhang Yichi (Catill Zhang) | Mobile No.: +65 89093764 | Email: zyc2068521267@gmail.com |
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EDUCATION

Nanyang Technological University (NTU), Singapore Aug 2023 - Jul 2027 (Expected)
Bachelor of Computing (Hons) in Data Science and Artificial Intelligence
• Honours (Distinction), current CGPA 4.45/5 .

Relevant modules:

SC2006 Software Engineering, MH3511 Data Analysis with Computer, SC4000 Machine Learning, SC4020 Data Mining

SKILL

Programming: Python (Pandas, NumPy, scikit-learn), C/C++, Java, JavaScript/TypeScript, HTML/CSS, Bash .
Machine Learning (ML)/Artificial Intelligence (AI): Transformers/Large Language Models (LLMs), MLLMs (Multimodal), Reinforcement, RAG/GraphRAG .
Data & Tools: SQL, MATLAB (basic), UiPath Automation .
Systems/HPC: Linux, PBS, OpenMPI basics .
Language: Proficient in English and Chinese .

EXPERIENCE

Multi-Modality Art Project (M-A-P) Organization Dec 2024 - Present
Research Intern

- Working with a top 10 Chinese AI open-source institution, driving large-scale AI projects and promoting an open-source community.
- Contributed to diverse LLM research projects, advancing areas such as GUI agents, math formalization, AI-driven creative writing, and auto-coding agents.
- Gained extensive experience in building benchmarks, data preparing and model performance evaluation.

NTU Inspire office Dec 2024 – Mar 2025
Prime Developer on Graphrag, Intern

- Developed and deployed a GraphRAG-based automation system for processing academic resources such as textbooks and past papers, streamlining course material preparation and enabling scalable rollout across 20+ NTU courses.
- Designed and implemented a semantic and metadata-driven architecture for sparse knowledge graphs, optimizing retrieval algorithms and achieving >90% retrieval rate in query success rate under targeted conditions.

PROJECT EXPERIENCE

NTU HippoCamp Sep 2025 - Nov 2025
Research Assistant

- Constructed the first comprehensive framework for assessing agents' ability to manage massive multimodal in formation spaces with personalized memory.
- Collecting multi-modality data including document, text, image, video, and audio files thus building a 42GB benchmark.
- This research have been submitted to CVPR2026.

Bytedance UI-TARS2 Dec 2024 - Sep 2025
Research Assistant

- Contributed to training an end-to-end multi-modality GUI agent, which achieved state-of-the-art performance on mainstream agent benchmarks such as Online-Mind2Web, AndroidWorld, and OSWorld as an open-source model.
- Served as the core data contributor, building a 690+-environment sandbox suite (including real HTML5/WebGL and LLM-generated environments), enabling scalable agent training with fully verifiable results.
- Reference: Hugging Face Papers — 2509.02544.

M.A.P CriticLean – Data Collecting Model-Design Mar 2025 - Jul 2025
Research Assistant

- Participated in establishing a critic-guided reinforcement learning framework, creating a data flywheel system to enhance the accuracy and robustness of mathematical formalization.
- Constructed a benchmark with over 40,000 high-quality mathematical formalization results.
- Paper submitted to ACL JULY; arXiv: 2507.06181 .

InputMagic - LLM prompt application GUI design

Jul 2024 - Aug 2024

Web developer and Prompt Engineering

- Constructed a sandbox-system on web, equipped with a LLM application enabling auto-coding, driven by natural language to change sandbox environment or adding novel functions. Over 100 people tried it rating high .
- Designed to combined LLM with CV tools to generate infinite creative contents. Highly praised by 8+ professors in NTU and UCB Link: <https://github.com/Yuba-Technology/InputMagic-Google> .
- Submitted to GOOGLE Gemini API Developer Competition App 2024 and received 20+ likes.
- <https://ai.google.dev/competition/projects/inputmagic> .

LEADERSHIP & CO-CURRICULAR ACTIVITIES

NTU High Performance Computer (HPC)

Feb 2025 - Present

Team Memeber

- Led a 4-person team for the **25th APAC AI HPC Challenge**, orchestrating 2-node (16×H100) DeepSeek inference on **Sglang**.
- Participated in the **Student Cluster Competition 2026**, contributing to achieving several state-of-the-art (SOTA) results.

Institution of Engineer and Technology (IET), Singapore

Aug 2024 - Present

NTU IET Logistic Head

- Leading a team of seven to support all kinds of back-up issues including dinner and venue reservation , cargo transport , items purchasing to serve in IET singapore with 200+ people in NTU.
- Conducted more than 12+ activities every year with 98% satisfaction rate.
- Received IET Certificate of Appreciation (2024): Recognition for contribution as Logistics Head for IET on campus.

NTU Vision of Hope Volunteering Club (VOH)

Sep 2023 – Sep 2024

Event Manager,Volunteer

- Weekly volunteering at Delta Sports Hall (SG), serving 12+ visually impaired participants by facilitating sports activities, ensuring safety, managing equipment, and designing inclusive games for participation; achieved 99% satisfaction.
- Organized 3 team bonding events, which were well-received by 13+ team members.